Matrix inequalities and norms

Schatten norm complement involving Lin's positive definite quantity

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale (historical): Extending the proved trace and Frobenius cases to all Schatten norms is challenging; the comparison of these particular means has specialist importance.

Resolution — 2026-09-11

Solved (affirmative). George Stepaniants's complete proof, **Theorem 1**, proves the displayed inequality for every dimension, every complex positive definite pair and every $1 \le p \le \infty$. The published Heron norm inequality and a positive matrix comparison give $\|A + B + 2G\|_p \le \|A + B + G + L\|_p$. Since $2(A + B + G + L) = (A + B + 2G) + (A + B + 2L)$, the triangle inequality then proves the desired comparison. Proof PDF · Standalone TeX.

The complete argument and its precise published input passed a separate Codex-agent review. It was developed with ChatGPT/Codex; the verification is independent agent review, not external human peer review or formal certification. Authorship, source checks, reproduction and public-branch audit. The contribution is the convexity deduction from existing comparisons. The original ID, path, statement, historical ratings and earlier status check below remain intact.

Problem statement

For positive definite $A, B \in \mathbb{C}^{n\times n}$, set

$$G = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2}, \qquad L = A^{1/2}(B^{1/2}A^{-1}B^{1/2})^{1/2}A^{1/2}.$$

Here $G = A\#B$, while L is the explicitly defined quantity used by Lin; no symmetry of $L(A, B)$ is assumed. Determine whether, for all $n \ge 1$, all positive definite A, B, and all $1 \le p \le \infty$,

$$\|A + B + G + L\|_p \le \|A + B + 2L\|_p.$$

For $p < \infty$, $\|X\|_p = (\operatorname{tr}|X|^p)^{1/p}$, where $|X| = (X^*X)^{1/2}$; for $p = \infty$, use the largest singular value.

Matrix means are central to interpolation of positive definite data. The conjecture supplies a norm comparison between two concrete sums of matrix functions; eigenvalue comparison between G and L alone is not the statement being requested.

References

1. M. M. Ghabries, H. Abbas, B. Mourad and A. Assi, *New log-majorization results concerning eigenvalues and singular values and a complement of a norm inequality*, preprint (2021), definition in §4, p. 11; Conjecture 4.1, p. 13. PDF; journal article.
2. M. M. Ghabries, *Contributions to Matrix Inequalities and Some Applications*, PhD thesis (2022), final Open Problems, Problem 5, pp. 109–110. Author-uploaded thesis.

Status check (2026-09-10): the first source proves $p = 1, 2$ and then conjectures the full interval. The thesis explicitly retains that gap. Current arXiv version v1, published metadata, the author's later July 2026 matrix-mean paper, and searches for "Ghabries Conjecture 4.1 norm", "Lin complement geometric mean Schatten", including 2025/2026, did not reveal a proof or counterexample for the full interval. No assertion of exhaustive coverage is made.